

Unified Graph: Invariant Visual Semantics

Abstract

This document presents the practical realization of Creative Information Theory (March 2026; DOI: <https://doi.org/10.5281/zenodo.19394773>), designed within the Invariant Visual Semantics engineering standard.

◇ *International Standard for Copyright Protection: Berne Convention Legacy (Since 1886)*

The Meta-Systems Design discipline embeds the principles of the Berne Convention into digital architecture, confirming the emerging copyright of digital art through Invariant Visual Semantics — a digital proof of authorial contribution.

The integration of ICU closes the gap in the provenance of the work, ensuring evidentiary continuity between the source and any of its embodiments (including creative synthesis environments). This preserves the authenticity of authorship when scaling the original work in cross-media environments (in games, cinema, the 3D industry, interactive and virtual worlds).

This standard under development is aimed at preserving cognitive energy, ensuring the privacy of the creative process, and securing the exclusive right of ownership over the results of intellectual activity.

◇ *Ontology of Intellectual Creation: A System for Actualizing Authorial Intent*

L-Protocol is a cognitive OS that crystallizes the processes of thinking and creation, translating them into the status of sociocultural realization. It is an engineering solution where the Art Producer is a Designer of Visual Ontologies — the source of synthesis for Intellectual Capital Units (ICU).

This invariant architecture unlocks the ontological patterns of adjacent fields (e.g., Creative Industries × Information Technology × Economics), forming a multidisciplinary bridge for their interaction and the capitalization of new information assets.

The 16-layer ontology serves as a logical environment for designing information systems within the Creative Industries, where human autonomy is preserved in the era of generative acceleration.

By enhancing intuition through a cognitive interface, L-Protocol visualizes **complex system dynamics**, rendering them transparent. This enables operations with data scales at the level of a Creative Systems Architect, transforming complexity into a manageable asset.

Intellectual capital dynamics is a non-linear process of value preservation and scaling over time (Open Aesthetic Temporal Cycle), in which accumulated cognitive experience is recursively compressed into high-density semantic invariants that are the foundation for the emission of new intellectual assets (ICU).

The realization and verification of this process within a distributed cognitive infrastructure is carried out based on the Temporal IP Provenance (TIPP) protocol.

◇ *Unified Graph: The Semantic Invariant of Digital Art*

Invariant Visual Semantics is a multidisciplinary engineering standard for copyright protection of a digital visual work, where the geometry of the work and its semantic essence are expressed within the logic of the Unified Graph.

The Unified Graph is an engineering specification of intent, serving as the semantic invariant of the work. If 2D Art is a phenomenon, then the Unified Graph is its digital genome. It captures the Lines of Force, Tension Nodes, and Chromatic Genome, uniting them into a digital map of connections with a precision that precludes double interpretation. Through a system of hashing and cross-referencing, each data block within the Unified Graph is anchored to a specific visual element.

The Chromatic Genome is a map of the distribution of key color arrays and their semantic weights, describing the geometry of color on the canvas by linking the palette with the Tension Nodes.

Semantic Security is a process of verifying the source invariant, where the discipline's authorial terminology is integrated directly into the Unified Graph data structure. This forms the "Semantic Anchor" architecture:

- The work is immune to unauthorized AI generations through the invariant semantics of the Unified Graph.
- The Unified Graph remains immutable through cross-verification with the work's architectonics.

Creative Engineering is a systems design methodology that combines engineering and technical knowledge with creative thinking to materialize an author's intent into practically viable products, technologies, and solutions.

Intellectual Capital Unit (ICU) is an indivisible intellectual asset where the Unified Graph and Digital Art coexist as a single, mutually reinforcing invariant.

◇ *Origin and Methodological Basis*

The Meta-Systems Design discipline (founded in 2025 by Dmitry Moiseev) is the originator of the Invariant Visual Semantics Protocol — the synchronization of Digital Art and Unified Graph, which unifies both geometric and semantic invariants.

A semantic invariant is the structure of an author's intent characterized by high Semantic Density, where every link constitutes an indispensable condition for the authenticity of the asset.

The methodological basis of the discipline encompasses cybernetics, systems theory, meta-design, and intellectual capital research.

The Invariant Visual Semantics engineering standard is formed on the foundation of a multi-year R&D program exploring the mechanics of intellectual capital. The architecture of the protocol integrates social sciences, the developmental experience of sociocultural ecosystems (City-Wide & Regional Initiatives), and the design of visual ontologies (Cognitive Design).

The L-Protocol captures the transition from the conceptual design of the meta-system and the creative synthesis engine (Syntengine Creatives, Dec 2023), through the design phase of feedback loops and **intellectual capital dynamics** research (Meta-Systems Design 'Research Phase', Dec 2025), to the stage of the practical synthesis of this capital's invariants (L-Protocol 'Formation Phase', March 2026).

◇ *Semantic Source*

The topologically complete 4x4 model represents a semantic processor. It initiates a recursive circulation of data, transforming external complexity into projected capital — spanning from raw input (L-01) to scalable ecosystems (L-16).

- LEVEL I — Cognitive Architecture (L-01 – L-03)
Internal capacities for pattern recognition and applied conceptual condensation
- LEVEL II — Knowledge Synthesis (L-04 – L-09)
Expression of insight into documents and frameworks that preserve and accumulate meaning
- LEVEL III — Knowledge in Practice (L-10 – L-12)
Application of experience within real-world constraints and adaptive feedback processes
 - EMERGENT ROLE — Cultural Producer Cognitive Capacity (L-13)
The unified creative capacity emerging from LEVELS I–III
- LEVEL IV — Knowledge Ecosystems (L-14 – L-16)
Circulation and development of concepts within collaborative and institutional networks

L-01	L-02	L-03	L-04
L-05	L-06	L-07	L-08
L-09	L-10	L-11	L-12
L-13	L-14	L-15	L-16

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